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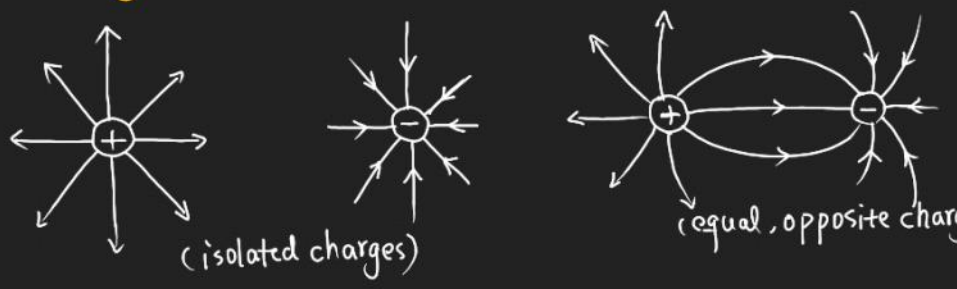
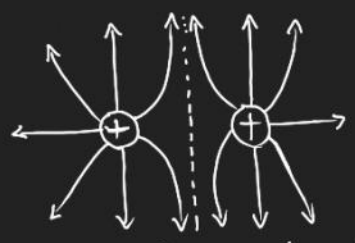
Electric field.

A region of space in which an electric charge experiences a force.



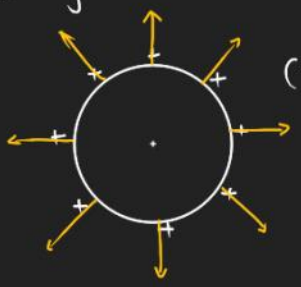


(equal, ~~same~~ like charges).

$E_A > E_B$

(E: electric field strength)

field lines are closer at A, so field strength is greater at A.



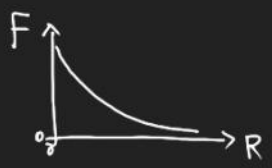
(spherical conductor).

Coulomb force between charges.

$$F = \left(\frac{1}{4\pi\varepsilon_0}\right)\frac{Q_1 Q_2}{R^2}$$ (R: separation between charges). $\left(F_G = G\frac{m_1 m_2}{R^2}\right)$

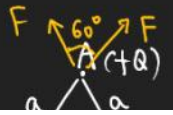
$\varepsilon_0 = 8.85\times10^{-12}\ C^2 N^{-1} m^{-2}$, $k = \frac{1}{4\pi\varepsilon_0} = 8.99\times10^{9}\ N m^2 C^{-2}$

(coulomb constant).

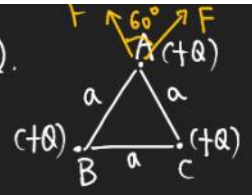


Q. Two ions, of charges $+3.2\times10^{-19}C$ and $-1.6\times10^{-19}C$, with a separation of 0.72 nm.

$$\frac{8.99\times10^{9}\times3.2\times10^{-19}\times1.6\times10^{-19}}{(0.72\times10^{-9})^2} = 8.9\times10^{-10}N.$$ attractive.

Q. 

Find the resultant force on charge at A.

Q. 

Find the resultant force on charge at A.

$F_{BA} = \frac{kQ^2}{a^2}$, $F_{CA} = \frac{kQ^2}{a^2}$

$F_A = \frac{kQ^2}{a^2} \times \cos 30^\circ \times 2 = \frac{\sqrt{3}kQ^2}{a^2}$.

Q.

A charge of +q moves on a straight line along +2Q and −Q.

Find a position at which the resultant force on +q is zero.

+q is placed in region C.

$\frac{kQq}{x^2} = \frac{k(2Q)q}{(d+x)^2}$, $\frac{1}{x^2} = \frac{2}{(d+x)^2}$

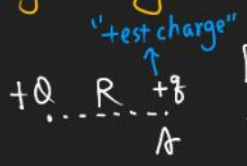
$\sqrt{2}\,x = d + x$, $x = \frac{d}{\sqrt{2}-1} = (\sqrt{2}+1)d$.

Electric field strength.

force per unit positive charge acting on a stationary point charge.

$\boxed{E = \frac{F}{q}}$ unit: $N \cdot C^{-1}$

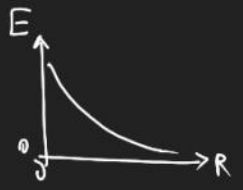
$\left(g = \frac{F}{m}\right)$



Electric field strength at position P produced by +Q

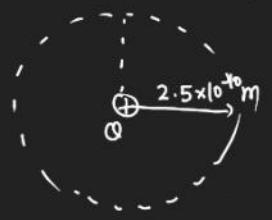
$E_A = \frac{kQq}{R^2} \Big/ q = \boxed{\frac{kQ}{R^2}}$

$\left(g = \frac{GM}{R^2}\right)$



Q. At a distance of 2.5×10^{-10} m from an isolated gold nucleus Au-197, which has 79 protons.

Calculate the electric field strength due to the nucleus.



$Q = 79 \times 1.6 \times 10^{-19}$ C.

$E = \frac{8.99 \times 10^9 \times 79 \times 1.6 \times 10^{-19}}{(2.5 \times 10^{-10})^2} = 1.8 \times 10^{12}\ N \cdot C^{-1}$.

Q.

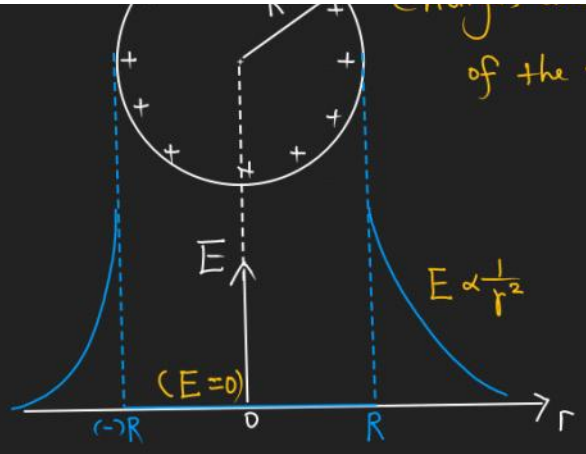
Find a position at which the resultant electric field strength is zero.

$\frac{k(2Q)}{(r-x)^2} = \frac{kQ}{x^2}$, $x = 0.4r$

$\frac{k(2Q)}{(r+x)^2} = \frac{kQ}{x^2}$, $x \approx 0.4r$.

Field of a charged sphere.

Charges are concentrated at the centre



of the sphere.

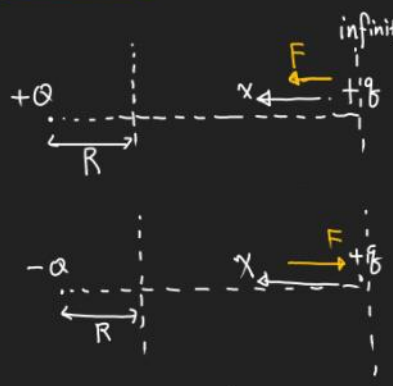
Q. An isolated metal sphere, of radius 21 cm, carrying a charge of 9.8×10^{-6} C.
Calculate the electric field strength on its surface.

$$E_{(surface)} = \frac{8.99\times10^{9}\times9.8\times10^{-6}}{0.21^2}$$
$$\approx 2.0\times10^{6}\ N\cdot C^{-1}.$$

Electric potential (V)

Work done per unit positive charge in bring a small test charge from infinity to the point.

$$\boxed{V = \frac{W}{q}} \quad (J\cdot C^{-1})$$

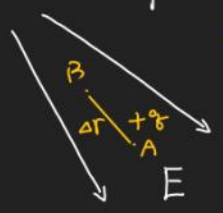


$$\boxed{V = \frac{kQ}{R}}$$

$$\boxed{V = -\frac{kQ}{R}}$$

$$\boxed{EPE = q\cdot V}$$

Relationship between electric potential and electric field strength.



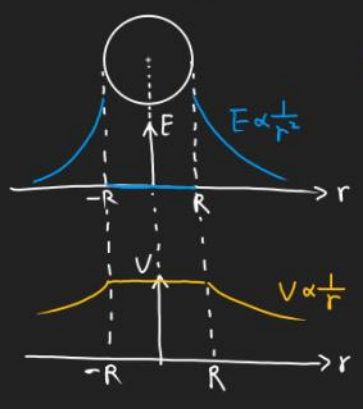
Δr is very small.
$E_A = E_B = E$.

$F = -F_E = -E\cdot q.$

$F\cdot\Delta r = q\cdot\Delta V_{AB}$

$-E\cdot q\cdot\Delta r = q\cdot\Delta V \quad \therefore E = -\frac{\Delta V}{\Delta r} \quad (-\frac{dV}{dr})$

electric field strength is equal to negative potential gradient.

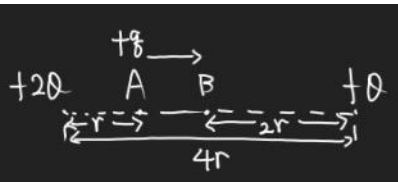


$$V = \frac{kQ}{r} \xrightarrow{E=\frac{dV}{dr}} E = \frac{kQ}{r^2}$$

Q.

Calculate the electric potential at the position A.

$$V_A = \frac{k\cdot2Q}{r} + \frac{k\cdot Q}{3r} = \frac{7}{3}\frac{kQ}{r}.$$



Calculate work done moving +q from A to B.

$W = \Delta EPE_{(AB)}$.

$$V_A = \frac{k \cdot 2Q}{r} + \frac{k \cdot (-Q)}{3r} = \frac{5}{3}\frac{kQ}{r}.$$

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